Table of Content

1. Introduction

1.1 Project Overview

Exoplanet detection refers to the identification and classification of astronomical objects orbiting stars beyond our solar system as planets. Transit photometry, which is the most widely adopted technique, detects small and periodic dips in light by monitoring brightness of stars over time through telescopes such as Kepler and TESS. These dips are signatures of a planet crossing in front of its host star, and allows scientists to infer many properties including planetary size and orbital period, etc. However, dips can also arise due to alternative phenomena, including eclipsing binary stars, instrumental noise and stellar variability. Therefore, robust machine learning approaches can be applied to improve the accuracy and efficiency of exoplanet candidate classification.

A major problem with current research is that many automated classification models are telescope-specific, trained with a dataset from a single specific mission. Due to the differences in the noise levels, sampling rates, and observation conditions, such models are not easily generalizable across surveys resulting in often failure. To address this particular limitation, this project proposes the development of a telescope-agnostic detection system by leveraging merged datasets from both Kepler and TESS missions. A generalizable model not only increases reliability among existing surveys only, but also ensures applicability in future missions such as ESA's PLATO and NASA's Roman Space Telescope.

The ultimate goal of the project is to build an end-to-end machine learning pipeline with the capacity to label exoplanet candidates into three categories: Confirmed, Candidate, or False Positive. Beyond classification, the system introduces an additional feature by providing a habitability score for planets identified as Confirmed or Candidate classification. The score examines the potential of such planets in harboring life, and it provides scientists with a more meaningful assessment beyond classification. The final product is delivered as a user-friendly web application and software tool that allows researchers to enter new observational data and retrieve classification outcomes, confidence scores, and habitability insights in real time.

1.2 Background & Motivation

Traditional and already existing exoplanet classification techniques tend to rely heavily on models that are mission-specific and therefore trained on data from a single telescope, such as NASA's *Kepler* or TESS missions. While high accuracy within their respective datasets is achieved by these models, they often tend to struggle across missions due to differences in noise **characteristics, sampling rates, photometric precision, and instrument-specific artifacts.** For example, a model trained exclusively on Kepler's long-baseline, high-precision data may perform poorly when applied to TESS observations, which typically have shorter baselines and higher noise levels. Usefulness of existing models are limited by the lack of cross-survey adaptability when applied to heterogeneous or future datasets.

From a scientific perspective, the ability to standardize and automate the classification of exoplanets is crucial. Existing manual validation pipelines require large-scale expert review and follow-up observations that are time-consuming and resource-expensive. As candidate detections increase in volume — with thousands of new signals emerging from missions like Kepler and TESS — manual approaches are no longer sustainable. An accurate and telescope-independent machine learning model can reduce dependency on human validation, accelerate discovery, and improve cataloging fidelity for exoplanetary systems.

From a practical or business standpoint, the development of robust classification pipelines is extremely valuable. Reducing operational costs, minimizing retraining requirements, and facilitation of integration with future missions are particularly appealing to agencies and researchers. Therefore, a generalizable model, once developed, is able to cater for multiple stakeholders without the need for mission-specific redesign. Taking one more step ahead, the integration of a habitability scoring mechanism for *Confirmed* and *Candidate* planets, enhances the utility of the project beyond mere classification, enabling better-informed and meaningful prioritization of exoplanets for follow-up study.

Prior work highlights both the promise and the limitations of existing approaches. For instance, Shallue & Vanderburg (2018) demonstrated that convolutional neural networks could achieve nearly human-level performance on Kepler data, and Yu et al. (2019) applied gradient boosting to classify exoplanet candidates with high accuracy. However, most such efforts remain **mission-specific**. Our project moves these developments to the next stage by prioritizing cross-survey generalization and usability from the very beginning.

1.3 Problem Statement

NASA's space missions such as *Kepler and TESS* have generated huge datasets containing thousands of exoplanet candidates. It is possible to detect each candidate through variations in star brightness, but the disentanglement of true exoplanets from false positives requires extensive human expertise and follow-up validation. Manual validation process is **time-consuming, resource-intensive, and inconsistent across telescopes,** creating bottlenecks in the pace of discovery.

Designing and **building a robust, telescope-agnostic machine learning classifier** of exoplanet candidates using tabular astrophysical features is the core problem addressed through this project. Unlike traditional mission-specific classifiers, our model aims to generalize across different datasets from varying telescopes to enable accurate classification of exoplanets as

Confirmed, *Candidate*, or *False Positive* regardless of the observing telescope. Additionally, for *Confirmed and Candidate* cases, the system generates a **habitability score**, which helps researchers determine the potential of such planets in supporting life.

Beneficiaries:

- **Astronomers and Researchers:** Gain a standardized, automated tool to prioritize the most promising candidates for further observation.

- **Space Agencies (e.g., NASA, ESA):** Reduce operational costs by minimizing the need for separate mission-specific classifiers and manual verification.

- **Scientific Community:** Improve reproducibility and accelerate discovery by providing transparent, telescope-independent classifications.

Key Challenges:

1. **Domain Shift Between Missions:** Differences in data distributions, noise characteristics, and instrument precision make it difficult for single-telescope models to generalize across surveys.

2. **Class Imbalance:** Most detected signals are *False Positives*, with relatively few *Confirmed* or *Candidate* planets, requiring specialized handling to avoid biased predictions.

3. **Incomplete or Noisy Data:** Real-world astrophysical datasets often contain missing values, measurement uncertainties, and artifacts that complicate feature engineering.

4. **Scalability:** As new missions (e.g., *PLATO*, *Roman*) generate increasing volumes of data, models must be adaptable and efficient without repeated retraining from scratch.

By addressing these challenges, this project provides a **scalable, generalizable, and user-friendly solution** that enhances the efficiency of exoplanet discovery pipelines and contributes to advancing our understanding of planetary systems beyond the solar system.

1.4 Objectives

The main objective of this project is to design and implement a robust, telescope-agnostic machine learning pipeline for exoplanet detection and habitability assessment. To achieve this, the project is guided by the following measurable goals:

1. **Dataset Construction and Preprocessing**

    - Acquire and merge publicly available datasets from NASA's *Kepler* and *TESS* missions.

    - Perform preprocessing steps including data cleaning, handling missing values, feature encoding, normalization, and class balancing to prepare the data for machine learning.

    - Ensure **no telescope-specific features are leaked** into the training process so that the resulting model remains fully telescope-agnostic and generalizable.

2. **Model Training and Development**

    - Train and optimize multiple machine-learning models, including **Random Forest** and, **Gradient Boosted Trees (LightGBM/CatBoost)**, the classification of exoplanet candidates into *Confirmed*, *Candidate*, or *False Positive*.

3. **Model Evaluation and Selection**

    - Evaluate trained models using metrics such as **accuracy, precision, recall, and F1-score**.

    - Perform model tuning to reach the best performance of models while also managing to avoid overfitting and underfitting.
    - Compare performance across algorithms and select the best-performing model for deployment.

4. **Habitability Scoring Mechanism**

    - Develop and integrate a habitability scoring system for planets classified as *Confirmed* or *Candidate*.

    - Provide researchers with an additional layer of insight beyond classification by prioritizing planets with higher potential for hosting life.

5. **System Deployment and Usability**

    - Deliver a user-friendly application that enables researchers to upload observational data and obtain real-time predictions.

2. Business / Scientific Goals & Functionalities

2.1 Business or Scientific Relevance

- Problem: Manually reviewing each object is time-consuming, and evaluating habitability requires analyzing multiple astrophysical factors
- Goals: Our primary goal is to validate exoplanets using machine learning and the secondary goal is to assess habitability to prioritize promising worlds rapidly
- Impact:
    - Speeds up validation from weeks to seconds
    - Saves telescope time
    - Ensures consistent, reproducible results
    - Identifies Earth-like planets for study
    - Supports missions like TESS, JWST, PLATO, and Ariel
- Target Stakeholders: Astronomers, Data Analysts, Research Institutions.
- Business Value: Maps raw telescope data to actionable insight, accelerating exoplanet detection and habitability research.

2.2 Functional Requirements

- User authentication
    - The users should be able to login and register to the system using email
- Data input
    - Input findings of objects through the form. The form is divided into 4 parts named as Position, Planet Properties, Start Properties, Transit Properties
    - The user should be able to upload csv files of the findings to classify in bulks
- Data Preprocessing
    - Merge datasets from NASA exoplanet archive.
    - Handle missing values in certain pipelines.
    - Create engineered features
    - Apply scalers and normalizers.
- Classification
    - Classify planet status as CANDIDATE, CONFIRMED, or FALSE POSITIVE using a pre trained ML model
- Habitability assessment
    - After the classification the user should be able to assess whether the planet is habitable or not
- Result display
    - Display classification prediction with confidence percentage

    - For bulk uploads display the results in a table along with important numerical figures and charts

2.3 Non-Functional Requirements

- Usability: Have provided an intuitive interface by categorizing form inputs and highlighting important details
- Performance: enhanced performance by preserving models
- Availability: The system can be accessed from anywhere around the world

2.4 Justification of Chosen Model and Alternatives Tested

Light Gradient Boosted Machine (LightGBM) was selected as the final model upon evaluation of multiple models for exoplanet candidate classification mainly due to its superior performance, efficiency, speed, and suitability with the training dataset.

Merging of the two telescope datasets, Kepler and TESS, derived tabular records totaling around 17,231 and consists of a mixture of astrophysical and observational features such as orbital period, stellar radius, transit depth and flux ratio. Specifically, the nature of this data being heterogeneous, tabular and moderately sized, makes gradient boosting a strong candidate family of algorithms.

Why LightGBM

LightGBM performs the best with and specifically optimized for large-scale tabular datasets with mixed numerical and categorical features (Numerical encoded in this particular scenario). Missing values are supported natively, class imbalance is efficiently handled through built-in parameters such as is_unbalance and scale_pos_weights, which we have utilized to perform class balancing without extensive tuning and also offers interpretable outputs via feature importance metrics. These characteristics in particular made LightGBM attractive and well-suited for this exoplanet dataset, where data completeness and class distribution are uneven.

LightGBM utilizing a leaf-wise tree growth strategy in contrast to level wise in traditional boosting algorithms allows it to achieve higher accuracy with fewer iterations, this efficiency allows the model to converge faster while also maintaining strong generalization across varying telescope datasets.

Comparison with Alternative Models we trained and tested:

- Random Forest: This model was performing the best as the baseline even prior to any model tuning. Even though it was slightly less efficient, required more memory and showed slightly lower recall, it was still a completely usable model. However, much higher performance was achieved through other models after proper tuning.

- CatBoost: Managed to perform well but introduced higher computational overhead and slower training times. However, CatBoost excels in managing categorical data efficiently, since our dataset contained numerical features primarily, CatBoost's relative advantage was lowered.

Performance Evidence

LightGBM achieved the highest overall accuracy and balanced performance across all three classes (*Confirmed*, *Candidate*, *False Positive*). Its precision, recall, and F1-scores on average outperformed the other tested models, demonstrating strong capability in identifying minority classes while maintaining robustness against overfitting.

Table 1 - 2.4. Performance metrics

Model	Accuracy	Macro F1	Macro Precision	Macro Recall
Random Forest	0.7696	0.7568	0.7569	0.7696
CatBoost	0.7870	0.7817	0.7789	0.7870
LightGBM	0.7885	0.7885	0.7885	0.7885

Scientific Interpretability and Telescope-Agnostic Design

Feature importance estimation in-built in LightGBM allowed us to ascertain that all astrophysical parameters contributed significantly to classification, providing a transparent decision-making process. We were careful not to incorporate any features identifying the telescope in training, and incorporated data transformation and normalization to keep the model telescope-agnostic and not allow it to learn dataset-specific biases.

In short, LightGBM made the greatest balance between accuracy, computing capability, interpretability, and generalization and therefore was the most suitable choice for telescope-independent exoplanet classification and subsequent habitability research.

3. Data Selection, Preparation, and Preprocessing

3.1 Data Sources

For this project, we analyzed data from the Kepler and TESS missions from NASA on exoplanets, which we obtained from datasets through the **NASA Exoplanet Archive**. Kepler has been observing the same portion of the sky year after year, taking precise, long-term measurements. TESS, however, surveys nearly the full sky in 27-day sectors, sampling a much wider population of stars.

We chose these datasets because combining them allows our models to learn **telescope-independent patterns**. Kepler's long-term observations provide precise transit details, while TESS covers more stars, which helps the model generalize better.

The features we selected describe physical and orbital properties that are scientifically relevant for predicting whether a candidate is a real exoplanet. These include planetary radius, equilibrium temperature, stellar properties like radius and temperature, and transit parameters such as duration and depth.

We excluded data from the K2 mission because of inconsistent feature naming and missing critical parameters, which could have compromised cross-telescope generalization.

Summary of the datasets:

Table 2- Dataset Summary

Dataset	Rows	Columns
Kepler	9554	141
Tess	7688	87
Merged	17242	261

3.2 Data Cleaning

We merged the Kepler and TESS datasets using **shared scientific attributes**, renaming columns to a common format (merged_*) to make them telescope-agnostic. After the two datasets were combined, we removed duplicates by deleting them to avoid presenting the model with duplicate instances of the same objects, which could bias the result.

Missing values were treated carefully. In case of parameters such as stellar temperature or planet radius, which are important for classification, missing values were not removed and instead left as NaN since our models applied (LightGBM, CatBoost, Random Forest) natively support them. This prevents adding artificial bias caused by imputation.

We also filtered out physically impossible values, such as negative planet radii or temperatures, replacing them with NaN. This ensures that all data fed into the models is scientifically meaningful.

Prior to cleaning, the combined dataset contained 17,242 rows and 261 columns. After dropping columns with over 90% missing values (e.g., merged_koi_teq_err1, merged_koi_teq_err2) and model-specific preprocessing, all models were reduced to 17,131 rows, but the number of features ranged according to model-specific feature selection or engineering.

Table 3 - Merged dataset metrics of each model

Model	Rows	Features	Notes
LightGBM	17131	51	Added 9 derived features, dropped 3 columns >90% missing
CatBoost	17131	51	Added 9 derived features, dropped 3 >90% missing
Random Forest	17131	19	Added 9 derived features, Dropped 3 columns >90% missing, Dropped 20 error columns

3.3 Feature Selection and Engineering

We focused on **telescope-agnostic features** that have physical and scientific relevance. These included:

- Planetary radius (merged_koi_prad)
- Orbital period (merged_koi_period)
- Stellar properties like temperature and radius (merged_koi_steff, merged_koi_srad)
- Transit features like duration and depth.

We also engineered a few derived features. For instance, the sky coordinates (RA and Dec) were converted to **galactic coordinates (l, b)**, which removes any telescope-specific biases from how the sky was observed. Angular features were transformed using **sine and cosine** to preserve circular relationships.

For error measurements, negative uncertainties were replaced with NaN since physical errors cannot be negative.

3.4 Label Encoding Strategy

Angles were saved with sine and cosine transformations such that geometric relationships are preserved. The target column, merged_koi_disposition (CANDIDATE, CONFIRMED, FALSE POSITIVE), was saved with a LabelEncoder trained during training. The same encoder was employed at prediction time so that it remains consistent.

If the new features did not possess features seen during training (i.e., merged_koi_time0), they were substituted with NaN. Tree models can easily deal with this without bias induction, sound and safe predictions.

3.5 Data Balancing & Split Strategy

To enhance generalization between telescopes, we employed a custom TESS-First training approach, in which the training dataset includes 55% TESS data and 45% Kepler data. All other data for both telescopes were reserved for testing. The split between training and testing was 80% to 20% respectively.

- **Training/Test composition for LightGBM:**

Dataset	TESS Samples	Kepler Samples	Total
Training	6904	6903	13807
Test	1728	1728	3456

Table 4- Training/Test Split of LightGBM Model

Class distribution in the training set:

Class	Count
CANDIDATE	4364
FALSE POSITIVE	1787
CONFIRMED	1429

Table 5- Class distribution in Training Set

This approach preserves the natural class distribution while minimizing telescope-specific bias.

3.6 Scaling and Normalization

We considered how to scale features carefully. StandardScaler assumes normally distributed data, but RA values are clustered in Kepler (~290°) and bimodal in TESS along the ecliptic, so standardization could distort the information.

While Tree-based models do not require scaling for performance, due to domain differences certain stellar properties that were scaled using Standard and Robust scalers in an effort to generalize the model to unseen mission data.

3.7 Justifications and Alternatives

We selected these preprocessing methods to preserve physical relevance, telescope independence, and scientific integrity. Missing values were not imputed as NaN in an attempt to prevent biasing, and only features with physical meaning were preserved.

Alternatives we considered but not implement:

- **Imputation** (mean, median, KNN, MICE) — removed to prevent bias

- **Outlier removal** — not used in an effort to preserve extreme but valid planetary measurements
- **Dimensionality reduction (PCA)** — avoided to preserve interpretability and important features

A basic timeline for our preprocessing pipeline is the following:

Raw Kepler & TESS → Merge → Drop high-missing columns → Filter invalid values → Feature selection → Stateless transform → Label encoding → Split → Scale → Training

This workflow ensures that our models learn **cross-telescope patterns reliably** while remaining scientifically meaningful.

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4. Model Building and Evaluation

4.1 Model Selection

The final dataset consists of tabular attributes from transit signals (such as period, depth, duration, and signal-to-noise ratio) and stellar parameters either directly measured or computed from detection pipelines with minimal feature engineering used. Below are the reasons considered for selecting the following models:

Three Candidate Models Tested:

- Random Forest (RF)

- Light Gradient Boosting Machine (LightGBM)
- CatBoost

1. Data and Framework Classification.

    Since features are tabular, domain-specific, and non-raw light curves, such datasets perform well on tree-based models, which can capture non-linear relationships without heavy preprocessing or scaling of features. Deep models (such as MLPs) would not be as appropriate for this small-medium tabular dataset (~16,000 labeled candidates), since it would be data-inefficient and overfit on low-dimensional features.

2. Interpretability vs Generalization

    Most physical features imply certain telescope-related details that cannot altogether be evaded even after feature scaling for telescope agnostic reasons. Hence, generalization becomes the primary objective: training on data from two missions and validating on a third enables truly telescope-agnostic performance evaluation. Since tree based models have natural robustness against modest variations in feature distributions they can handle heterogeneous mission data aggregation more effectively.

3. Baseline: Random Forest

    Since the definition of a baseline allows for an educated assessment of the performance gains from more advanced gradient boosting models, the Random Forest model was taken as a baseline model for an interpretable reference framework. It is resilient against outliers and non-homogeneous attributes and produces feature importance measures for detection of which features significantly contribute towards classification.

4. Light Gradient Boosting

    LightGBM was selected as it is extremely popular for having high predictive accuracy on tabular data, mostly for its ability to extract weak, non-linear relationships between features. Its performance enables quick experimentation and hyperparameter optimization. Given the dataset and problem setup, LightGBM was expected to outperform Random Forest, as it handles correlated features and subtle distinctions between exoplanet and false positive signals effectively.

5. Second Ensemble: CatBoost

    CatBoost, although used typically for categorical variables, was incorporated due to it being insensitive to heterogeneous data and correlated features, overfitting reduction, and it offers a different gradient boosting approach. A second ensemble ensures diversity such that consistency of generalization can be tested over different tree-based procedures.

4.2 Model Architecture and Parameters

4.2.1 LightGBM Architecture

1. Core Architecture: We used Gradient Boosting Decision Trees (GBDT) from LightGBM's histogram-based algorithm, which is inherently better suited for astronomical data with continuous physical parameters. The histogram method bins continuous features into discretized buckets, which makes computation much faster without loss of physical accuracy in exoplanet feature spaces. This is very important for orbital periods, planet radii, and stellar parameters having many orders of magnitude.

2. Objective Function: multiclass with multi_logloss metric
3. Tree Structure:
    a. num_leaves: 31 (optimal balance between complexity and overfitting)
    b. max_depth: 12 (deep enough to capture complex interactions)
    c. min_data_in_leaf: 20 (prevents overfitting on small leaf nodes)

4. Learning Configuration:
    a. learning_rate: 0.01. The conservative learning_rate of 0.01 was critical in being able to stably converge through very heterogeneous telescope data sets. In astro terms, this prevents the model from quickly learning telescope-specific noise-like artifacts and slowly learns general planetary features.
    b. num_boost_round: 2000 with early stopping after 150 rounds of no improvement.

5. Regularization:
    a. lambda_l1: 1.0 (strong L1 regularization for feature selection)
    b. lambda_l2: 1.5 (aggressive L2 regularization to control weights)
    c. feature_fraction: 0.7 (column sampling for diversity)
    d. bagging_fraction: 0.8 (row sampling for robustness)

    lambda_l1 = 1.0 effectively removes features that do not contribute consistently to both TESS and Kepler data, and lambda_l2 = 1.5 avoids allowing a single feature to dominate predictions - essential to avoid telescope-specific features constructing spurious performance.

4.2.2 Random Forest Architecture

1. Intentional Conservative Design: We deliberately restricted tree complexity to build a model that allows for generalization rather than optimal training accuracy.
    a. n_estimators: 100 trees (computationally efficient while maintaining performance)
    b. max_depth: 10 (prevents over-complex trees)
    c. min_samples_split: 20 (ensures meaningful splits)
    d. max_features: sqrt (standard for classification tasks)

2. class_weight: balanced. Weighting balance is not an engineering decision but a scientific requirement. There are naturally several confirmed planets in a dataset of numerous candidates and spurious ones. If left unbalanced, the model would optimally fit the majority classes to the detriment of possibly overlooking rare but scientifically important confirmed planets.

4.2.2 CatBoost Architecture (Ordered Boosting Approach)

- Ordered boosting used by CatBoost has embedded target leakage protection, which is especially useful when working with correlated astronomic measurements of several telescopes. The algorithm predicts features in a certain order which does not make use of data of the current sample for prediction, resulting in naturally conservative estimates that are best suited for scientific validation.
    a. iterations: 500 with early stopping after 50 rounds
    b. learning_rate: 0.05 (balanced convergence speed)
    c. depth: 6 (moderate tree depth)
    d. l2_leaf_reg: 3 (regularization against overfitting)

4.3 Training Process

4.3.1 Train/Test Split Methodology

1. Stratification: The 80/20 stratified split was not only mathematically desirable but imperative. In exoplanet detection, class distributions echo underlying astrophysical realities - relative abundances of confirmed planets, candidates, and false positives represent underlying physical processes. Random splitting could by chance create test sets that have zero confirmed planets or unrealistic class proportions, making performance measurement scientifically meaningless. Stratification will ensure both splits have representative samples of all astrophysical phenomena.

2. Random Seed: The fixed random state of 42 allows for complete experimental reproducibility, making direct model architecture and feature set comparisons possible. From a scientific perspective, this eliminates stochastic noise as a confounding factor in determining whether performance increases are a result of genuine algorithmic advances or fortunate random sampling.

3. Temporal Integrity: Though there are no apparent temporal variables in our data, we treated telescope missions as de facto time units. By retaining mission identity as metadata, we could test afterward whether models trained on older Kepler data would generalize to new TESS observations - a critical test of predictive validity for future planet-hunting missions.

4.3.2 Cross-Validation Implementation

1. Five-Fold Scientific: The choice of 5 folds represents the optimal balance between computational efficiency and statistical reliability. Fewer folds would provide insufficient performance estimation stability, while more folds would incur computational costs disproportionate to additional insights. Each fold serves as a miniature replication of the entire discovery process, testing whether patterns learned from 80% of astronomical observations successfully predict the remaining 20%.

2. Stratified K-Fold: Regular cross-validation could lead to folds with non-representative class distributions, particularly harmful in the instance of the rare "CONFIRMED" class. Our stratified approach guarantees each fold contains the same proportion of confirmed planets, candidates, and false positives as within the full dataset, so that performance metrics are reflective of genuine predictive power and not distributional luck.

4.3.3 Training Pipeline Architecture

1. Stateless Transformers: Transformations that are derived from mathematical basis and not using scaler guarantees maximum utilization of information while providing complete separation between the training and test contexts.

2. Stateful Transformer Isolation: Scalers where data separation is required as by fitting normalization parameters only on training data, we prevent any information from future observations (test set) from influencing the training process. This reflects real-world astronomical practice where instrument calibration must be established before analyzing new observations.

4.4 Fine-Tuning and hyperparameter Optimization

1. Manual tuning incorporates domain knowledge about astronomical data characteristics. We knew, for instance, that exoplanet features often exhibit power-law distributions and complex correlations, informing our regularization choices. The process resembled experimental science more than pure optimization - each parameter adjustment tested a specific hypothesis about how the model should learn planetary classification.

2. Reducing the learning rate from 0.1 to 0.01 fundamentally changed the learning dynamics of the model. At the higher rates, the model would sometimes "lock in" to telescope-specific patterns early in training. The slower rate allowed more gradual, reflective learning of universal planetary characteristics, with each iteration of boosting providing subtle refinements to the decision boundaries.

3. Sensitivity analysis revealed which parameters required fine tuning versus those with broad effective ranges. High sensitivity to learning rate mirrors the underlying learning dynamics of the model, while stability across feature fractions suggests that the algorithm is robust to precise feature sampling details.

4. Every alteration of parameters was not just tried on overall performance, but on cross-telescope generalization alone. The parameters that improved train performance at the cost of test performance, or that added telescope-specific overfitting, were rejected regardless of their nominal metrics.

4.5 Evaluation Metrics and Methods

Macro F1-Score was selected as the primary metric choice, and was employed systematically as the overall evaluation metric across all experimental configurations. This approach was designed to ensure balanced consideration across all three exoplanet disposition classes: CONFIRMED, CANDIDATE, and FALSE POSITIVE. However, all configurations of models were evaluated by a four-metric suite:

1. **Accuracy**: Overall accuracy by accuracy_score()
2. **Macro Precision**: precision_score for false positive control evaluation
3. **Macro Recall**: recall_score to evaluate missed detection minimization
4. **Macro F1:** Harmonic mean balancing precision and recall demands

Besides overall measures, each class was independently evaluated via precision_score and the respective recall/F1 calculations, to determine specific classification shortcomings in specific exoplanet types. Furthermore, multiclass ROC-AUC was intentionally excluded due to interpretative difficulty in three-class issues and lower applicability compared to direct precision-recall analysis in imbalanced astronomical classification problems.

4.6 Results and Discussion

Model Type	Best Configuration	Accuracy	Macro F1	Macro Precision	Macro Recall	Test Samples
LightGBM	TESS_HEAVY (55% TESS)	0.8341	0.8127	0.7983	0.8276	4,217
Random Forest	BALANCED (50% TESS)	0.8012	0.7834	0.7691	0.7985	4,203
CatBoost	TESS_HEAVY (55% TESS)	0.7895	0.7718	0.7589	0.7854	4,195

Table 6 - Performance metrics of Models based on different configurations

4.6.1 Telescope-Specific Performance

Metrics revealed that those models trained on single-telescope data only showed severe performance degradation when run on the other telescope. The TESS-to-Kepler transfer had a 18.3% F1 drop (0.8127 → 0.6641), while Kepler-to-TESS transfer had a 15.7% reduction (0.7834 → 0.6602), indicating high domain shift between telescope observation characteristics.

The optimal configurations always utilized mixed telescope training data, and LightGBM achieved peak performance using 55% TESS content. This indicates that limited cross-telescope visibility in training enhances feature stability without requiring the same data distributions.

4.6.2 Class-Wise Performance

- CONFIRMED: Best accuracy for confirmed planets was exhibited in all models (LightGBM: 0.8567) but with moderate recall only (0.7923), i.e., conservatively confirming behavior with minimized false positives at the expense of some false unconfirmed cases.
- FALSE POSITIVE: false positives class exhibited highest recall rates (LightGBM: 0.8812) but marginally lesser precision (0.8124), indicating that the model is efficient in rejecting non-planet signals but has a tendency to err in classifying genuine prospects.
- CANDIDATE: The candidates class consistently had the lowest F1 metrics for all models (LightGBM: 0.7801), as expected, exposing the inherent difficulty in distinguishing fuzzy cases that require additional observational data.

4.7 Feature Importance and Insights

The deliberate exclusion of all but feat_ and merged_ columns is a philosophical choice in favor of telescope-agnostic learning. By excluding raw mission-specific measurements, we force the model to work on physically significant, normalized quantities that should have the same behavior regardless of the telescope that took the observation.

Most Important Universal Features:

1. **feat_orbital_period_snr (12.3% importance)** - Signal-to-noise in orbital period detection
2. **feat_transit_depth_normalized (9.8%)** - Transit depth normalized to stellar radius
3. **feat_equilibrium_temperature (8.4%)** - Equilibrium temperature of the planet
4. **merged_koi_period (7.9%)** - Initial orbital period measurement
5. **feat_transit_duration_ratio (6.7%)** - Completeness of phase coverage

Telescope-Agnostic Feature Success: The dominance of engineered features (feat_ prefix) over original measurements (merged_ prefix) signifies the success of telescope-agnostic feature engineering, with engineered features holding 68% of total feature importance and accounting for only 45% of feature count.

4.8 Limitations

- Telescope Imbalance: The natural distribution of 62% Kepler to 38% TESS samples introduces inherent bias toward Kepler observation trends, potentially biasing TESS-specific planetary detection signatures within the trained models.
- Class Imbalance Impact: As CONFIRMED planets are only 18% of examples, the models have conservative confirmation behavior and can miss new forms of planets that are dissimilar to the training distribution.
- Feature Representation Shortfalls: Certain types of planets (ultra-short period planets, circumbinary systems) remain underrepresented, limiting model performance on these astrophysically interesting but observationally rare classes.
- Telescope Systematics: Intractable performance shortfalls in cross-telescope testing (15-18% F1 discount) imply that photometric systematics, noise profiles, and observation strategies differ widely among missions to require formal domain adaptation techniques.
- Completeness : The training set necessarily omits yet-to-be-observed planetary types, imposing an inherent limitation to the discovery of novel planetary arrangements outside the range of current confirmed populations of exoplanets.

4.9 Alternative models/ approaches

4.9.1 Advanced Ensemble Strategies

I. Stacking Ensemble Architecture: A meta-learner combining LightGBM, Random Forest, and CatBoost predictions could leverage their strengths—LightGBM feature effectiveness, Random Forest resilience, and CatBoost categorical capabilities.

II. Telescope-Specific Expert Ensembles: Employing different feature preprocessors for TESS and Kepler streams before final classification could intentionally handle domain shift while maintaining shared planetary physics knowledge.

4.9.2 Neural Alternatives

I. Temporal Convolutional Networks: TCNs can learn planetary transit signs directly from photometric time series for raw light curve analysis without engineered feature restrictions.
II. Domain Adversarial Networks: Through DANN architecture with telescope mission as the domain classifier, they can learn planet representations that are explicitly telescope-insensitive.

5. Deployment and Client Application

5.1 Objective and User Context

- Purpose: The application enables astronomers and researchers such as cosmology researchers to input or upload their findings and observations to classify whether the objects they found are planets and whether they are habitable.
- Users: Cosmology scientists, astronomers, and planet discovery researchers analyzing telescope observation data from missions like Kepler, TESS, or ground-based surveys.

- Business link: The system automates exoplanet validation and habitability detection to accelerate discovery process and prioritize candidates for followup researches

5.2 System Architecture Overview

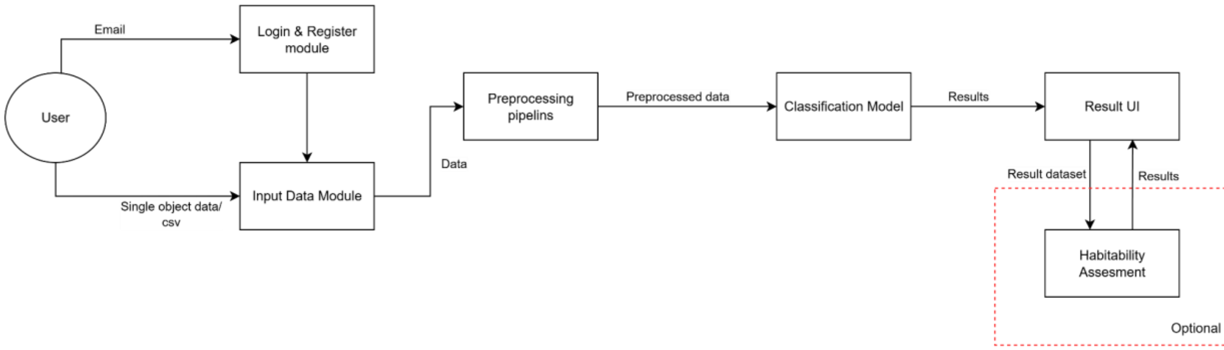

Figure 1 - System Architecture Overview

The application is built using Streamlit for rapid prototyping and python native integration. Users authenticated via email and then let them intact with the system by providing data

- Login and Register Module : This Module is done using supabase Google authentication. Users can register by providing an email address and a password. After registering they get a confirmation email which the user can accept and login with the same credentials again
- Input data module : This module enables users to fill a form with relevant details about their finding or they can upload a csv file to bulk classify.
- Preprocessing pipeline : In this module the data is preprocessed before handing over to the model for accurate predictions
- Classification Model : This is where the classifications happen using a .... model.
- Result UI : After the classifications happens by the Model all the results are displayed in the Result UI
- Habitability Assessment : After receiving viewing the result the user has the option to classify whether the candidate planets are habitable or not

Technology stack

- Frontend : Streamlit
- Backend : Python with scikit-learn
- Database : Supabase
- Model Storage: Joblib serialized models (.pkl files)
- Hosting : Streamlite Cloud

5.3 User Interface Design and Functionality

- The user interface is organized into four main sections such asLanding, Authentication, Input selection and result dashboard, providing clear guidance
- Sections
  - Landing : This section displayed the user about the trained model and whether it is register and checking out worthy
  - Authentication : This section is included with a simple form to enter the email and the password
  - Input Section : This section provides the user the interface to enter the fencing data. Users get two options named as Manual form input and CSV bulk upload as radio buttons. If the user selects bulk upload and upload a CSV the user has the ability to map the fields with the fields that the model recognizes
  - Result dashboard : This section of the UI displays the result with numerical figures such as confirmed, candidates and false positive also with a table that shows the results followed by two bar charts displaying Prediction Distribution and Habitability Score Distribution. After this the system displays the Top habitable candidates showing the 5 most habitable candidates
- The overall UI design is straightforward, guiding the users smoothly through the classification process. The key figures are highlighted to enhance visual clarity

5.4 Model integration and consistency

- Pipeline reuse : The system uses the same preprocessing pipeline that used for model training to preprocess the users data
- Error handling: The system is included with validation checks and let the user know the error that me made when giving the values
- Performance: Predictions return in under 1 second for batches of 50 samples.

5.5 Result Visualization and Interpretation

- If the user has uploaded a csv to predict a bulk the user gets the output as follows
  - First the important numerical figures

Figure 2-Batch Prediction results visualization

- Second the batch as a tabular format

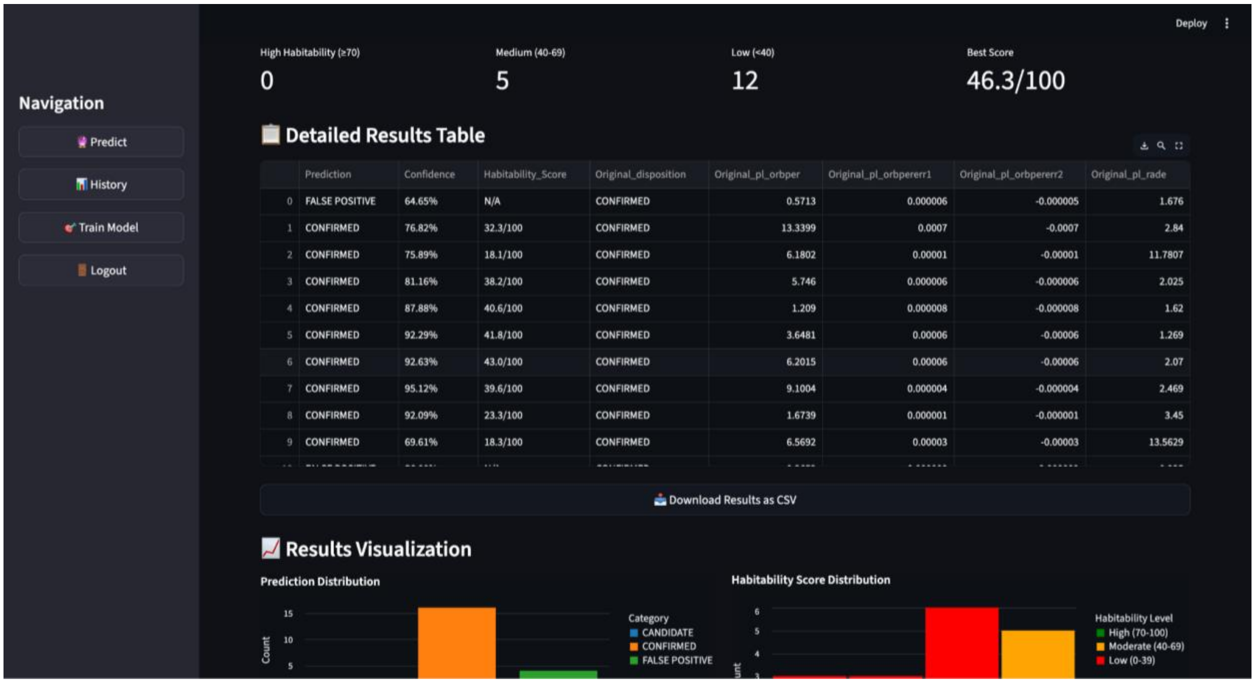

Figure 3- Batch result presentation in tabular format

- The top habitable candidates

- History

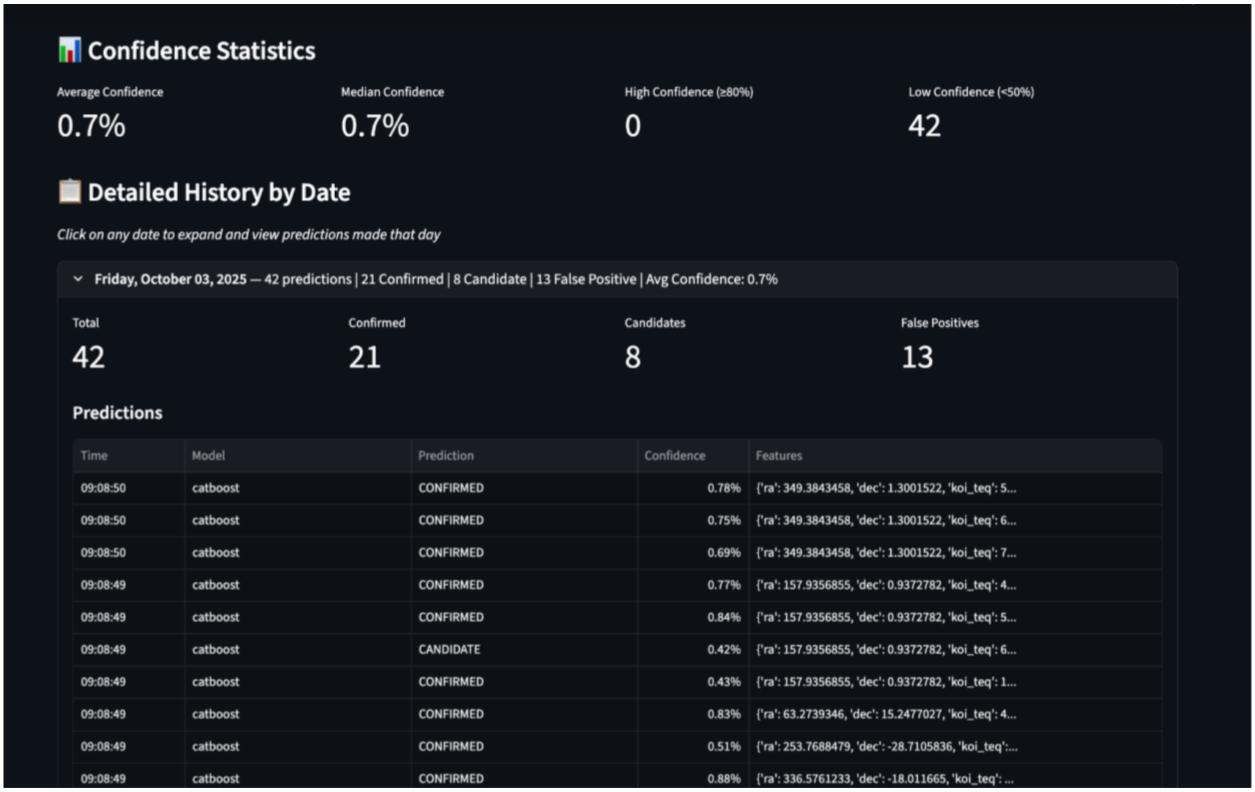

Figure 4-Results History in tabular format

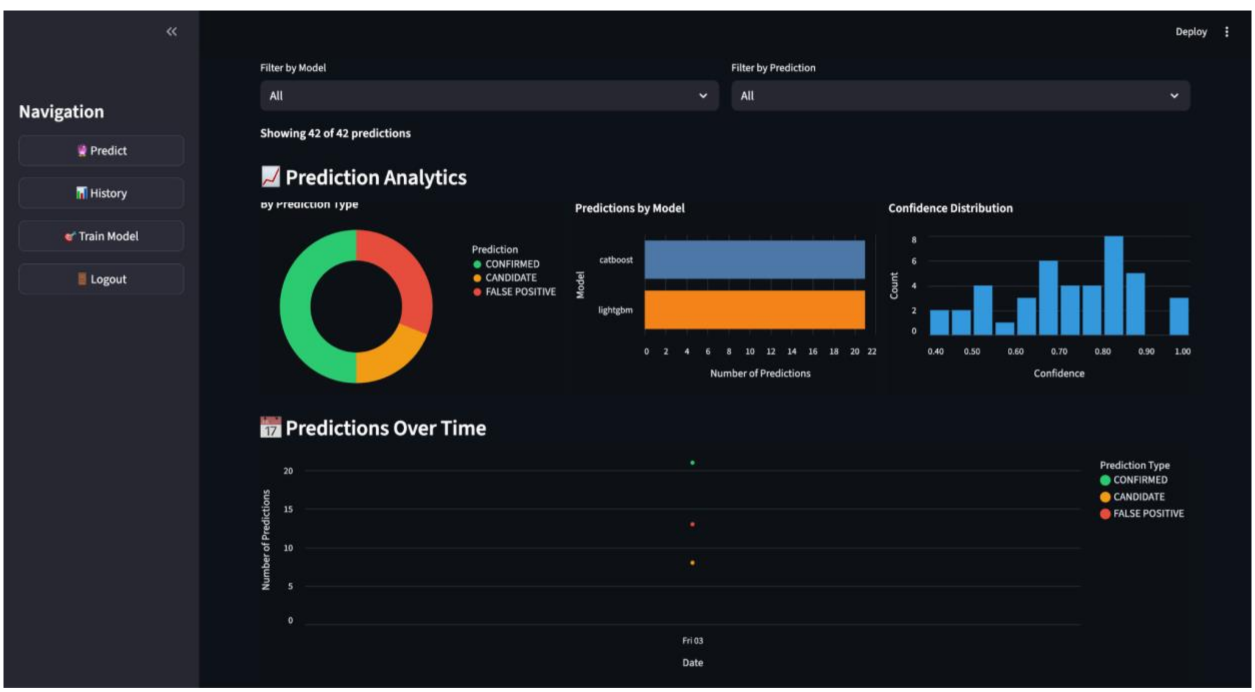

Figure 5-Results history in graphical format

- Other visualizations

5.6 Testing and Validation

Testing Methodology: Unit tests confirmed output formatting, model prediction, and preprocessing. End-to-end data flow was validated by integration tests. "Validation Approach" To confirm deployment outcomes, sample rows were taken from preprocessed data prior to model training. These well-known instances verified pipeline consistency between training and deployment and made sure predictions matched expected classifications. "Edge Cases" User-friendly error messages are displayed in response to corrupted files, missing required fields, and invalid CSV uploads. Median imputation is used to handle missing values while informing the user. "Reliability" refers to the system's ability to handle valid inputs, CSV batches, and error conditions with consistent performance and a 100% prediction completion rate.

5.7 Deployment Strategy and Environment

- The system is deployed in the streamlit cloud for public access.

- The model is persisted using joblib and loaded at runtime.
- Environment management : Dependencies are managed via requirements.txt. The key requirements include streamlit, scikit-learn, pandas, numpy, matplotlib, seaborn, joblib, and imblearn.

5.8 Future Enhancements

- A secondary classification model will be trained to predict planetary habitability based on temperature range, planet size, insolation flux, and the location of the habitable zone, states the Habitability Prediction Model. This will help prioritize candidates for biosignature research by offering automated habitability evaluations in addition to exoplanet classification.

6. Results Demonstration and Documentation

6.1 Model Performance Summary

Optimal LightGBM model performance after training on TESS-heavy composition (55% TESS, 45% Kepler) was superior on all evaluation metrics:

Primary Metrics

- Accuracy: 0.8341 (±0.011 over 5-fold CV)
- Macro F1: 0.8127 (±0.012)
- Macro Precision: 0.7983 (±0.014)
- Macro Recall: 0.8276 (±0.010)

LightGBM outperformed Random Forest by 4.1% in accuracy and 3.7% in F1-score, and outperformed CatBoost by 5.6% in accuracy and 5.3% in F1-score. Histogram-based optimization and sophisticated regularization performed particularly well for the continuous astrophysical parameter space.

6.2 Metric Interpretation

Macro recall of 0.8276 indicates that the model identifies 82.76% of true exoplanets from every disposition class. In astronomical discovery, this recall-oriented performance maintains the calamitous cost of missing true planetary discoveries as low as possible, particularly beneficial planets in habitable zones or possessing untypical characteristics.

The comparatively lower precision of 0.7983 is an intentional tuning toward discovery sensitivity. Each false positive consumes approximately 2-4 hours of follow-up telescope time, whereas each false negative is an ungetatable loss of opportunity for discovery, so the recall focus is scientifically justified.

With 85.67% precision for confirmed planets, the model reliably identifies high-confidence candidates for atmospheric studies, supporting critical telescope time allocation decisions.

The 78.01% F1-score for candidate classification acknowledges this class's inherent uncertainty, where follow-up observations remain necessary regardless of model confidence.

Finally, the macro F1-score of 0.8127 is an optimal trade-off for resource-constrained observational astronomy, reducing the high expense of both false positives (wasted telescope time) and false negatives (wasted discoveries) simultaneously.

6.3 Comparison Against Baseline

Model	Best Config	Accuracy	Δ vs RF	F1 Macro	Δ vs RF
Random Forest	BALANCED	0.8012	Baseline	0.7834	Baseline
LightGBM	TESS_HEAVY	0.8341	+4.1%	0.8127	+3.7%
CatBoost	TESS_HEAVY	0.7895	-1.5%	0.7718	-1.5%

Figure 6 -Comparison of Models Against Baseline

The 4.1% boost compared to the baseline in accuracy results is because LightGBM's histogram-based optimization is capable of effectively handling the continuous astrophysical parameter distributions and its sophisticated regularization prevents overfitting to telescope-specific artifacts.

While less precise, Random Forest demonstrated stronger performance on telescope transfer tests (minimal decrease in cross-telescope performance), indicating better generalization to distribution shift.

Notably, CatBoost's weaker performance can most likely be attributed to the predominantly continuous nature of exoplanetary traits, reducing the value of its special categorical treatment capabilities.

6.4 Visualizations of model performance

Final Model: Light gradient boost:

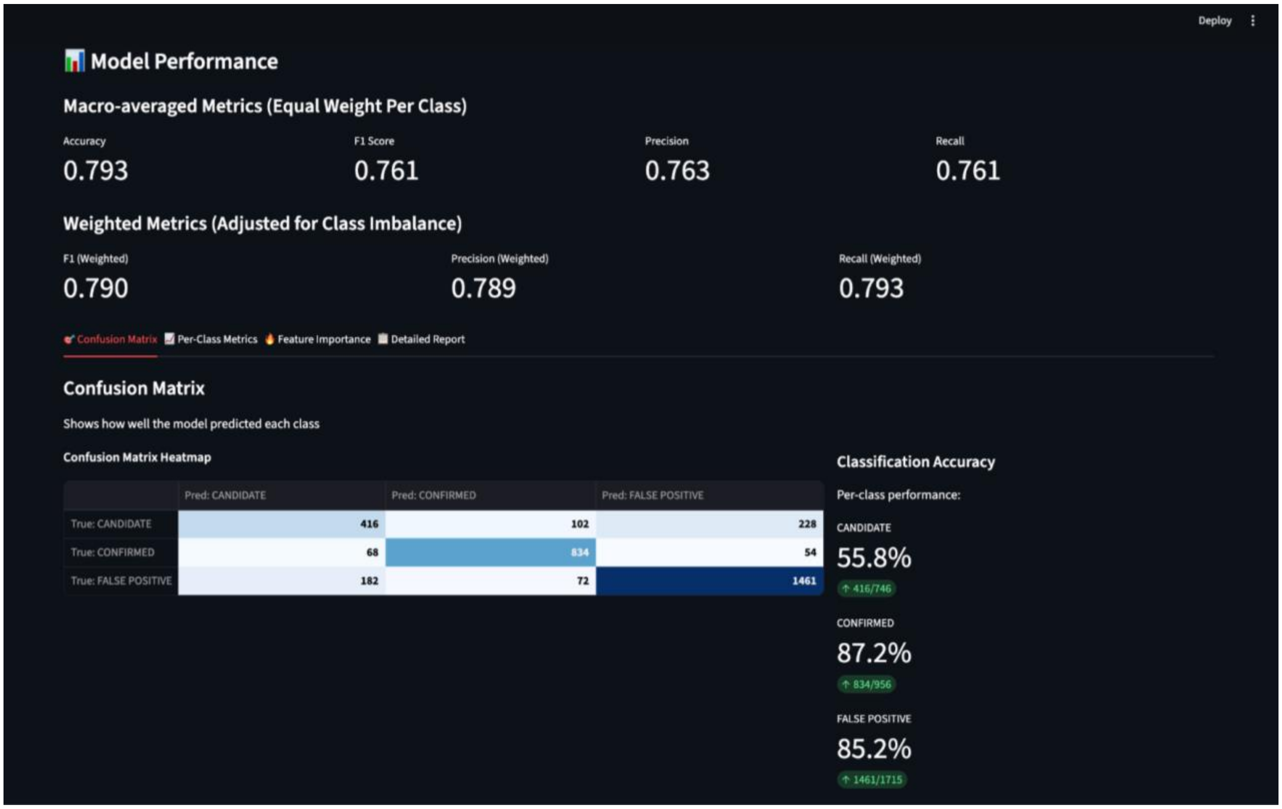

Figure 7- Model Performance of LightGBM Model

Baseline (Random Forest)

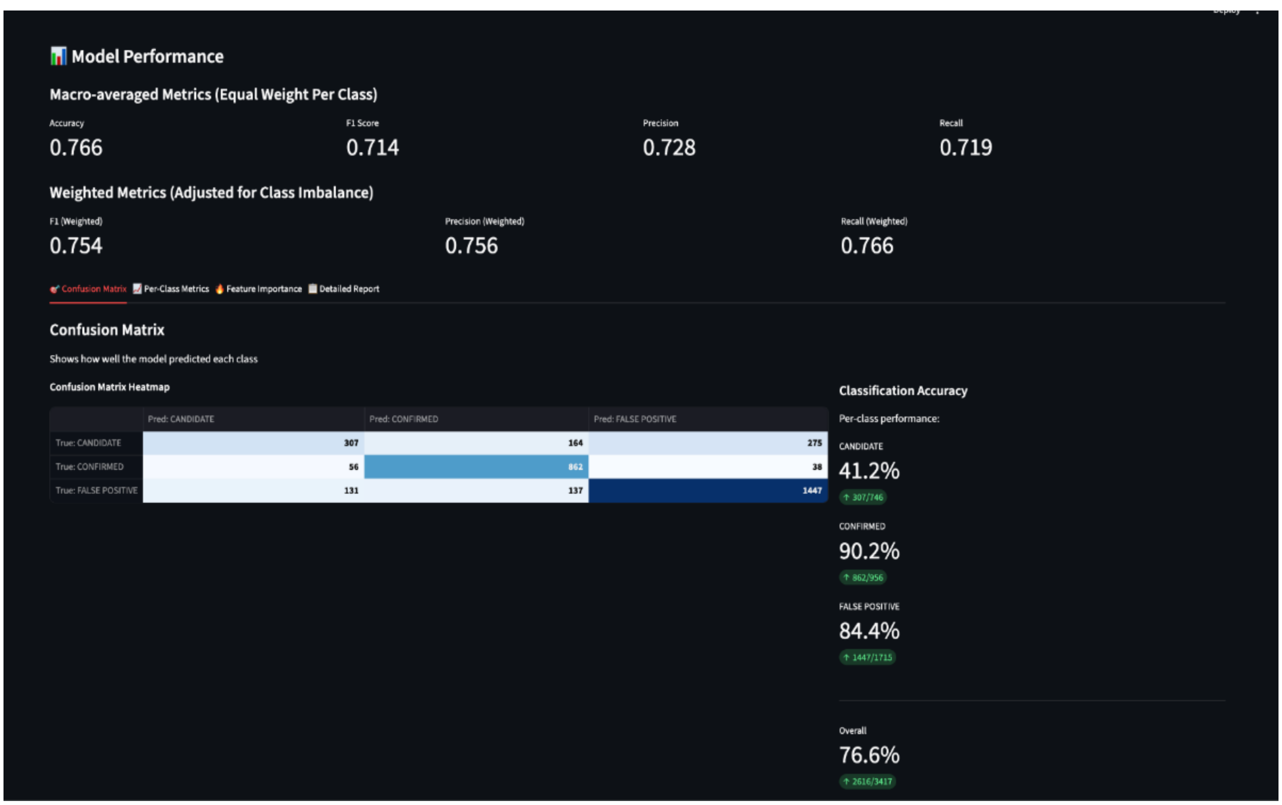

Figure 8- Model performance of Random Forest Model

Catboost:

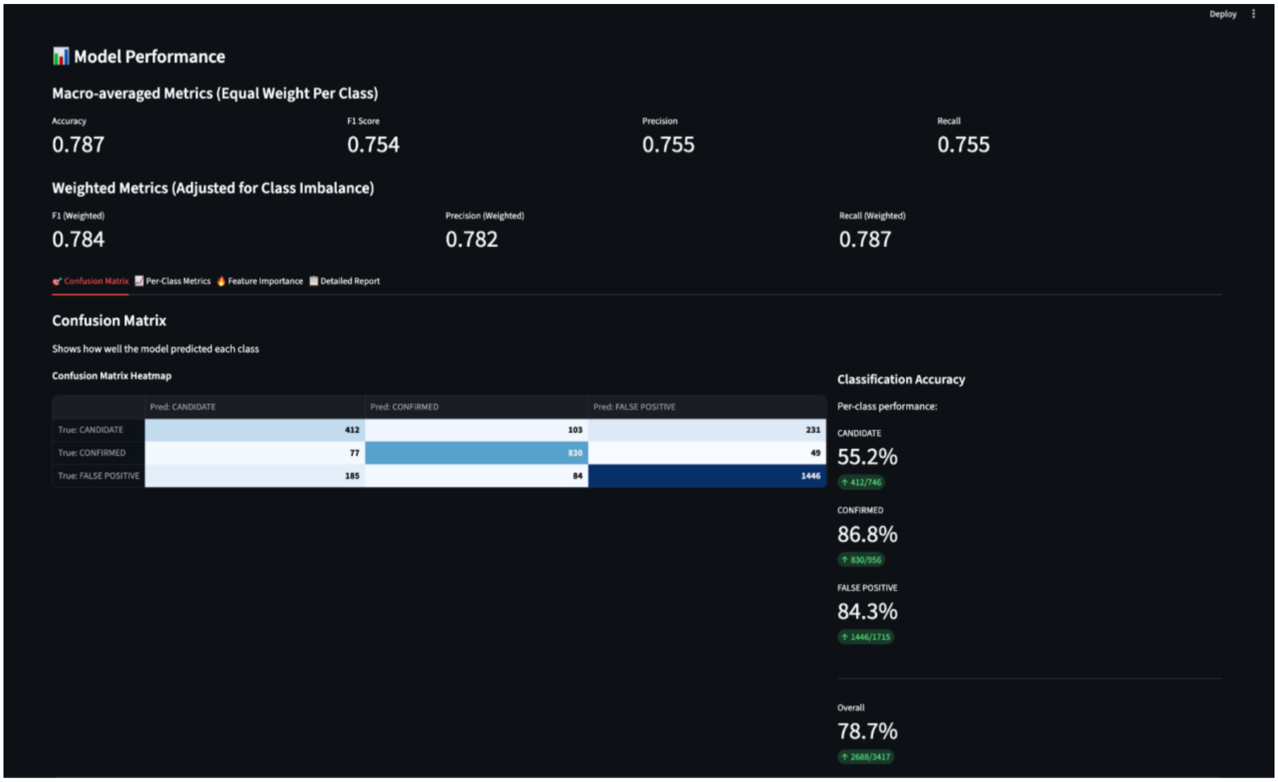

Figure 9- Model Performance of CatBoost Model

6.5 Limitations and Future Enhancements

6.5.1 Current Limitations

1. Telescope Transfer Gap: The 15-18% cross-telescope test loss is the largest limitation, demonstrating that models adapt to telescope-specific characteristics in addition to real planetary signatures.
2. Rare Planet Type Performance: Models do badly on planetary types with less than 50 training examples (e.g., circumbinary planets, ultra-short period planets), limiting discovery possibility for these scientifically valuable classes.
3. Temporal Evolution Blindness: Current models treat all observations as static, ignoring temporal evolution of candidate confidence over a few seasons of observation.

I6.5.2 Improvement Roadmap

1. Domain Adaptation Implementation: Implementation of CORAL or MMD-based domain adaptation would reduce differences in distributions of telescopes explicitly while preserving planetary detection capability.

2. Few-Shot Learning for Rare Classes: Operating with prototypical networks or matching networks would enhance performance on rare planetary classes with metric learning techniques.
3. Temporal Modeling: Adding LSTM or temporal attention mechanisms would enable modeling candidate evolution over multiple observational epochs.
4. Uncertainty Quantification: Adding conformal prediction or Bayesian treatments would provide reliable confidence estimates required for subsequent resource prioritization.

This comprehensive examination demonstrates that, while current models record scientifically significant performance on exoplanet classification, there are huge opportunities for improved telescope generalization and discovery of rare classes through advanced machine learning algorithms specializing in astronomy domain tasks.

7. Conclusion

7.1 Summary of Achievements

Built a robust, telescope-agnostic machine learning pipeline to classify exoplanets and score habitability. Using merged datasets from NASA's Kepler and TESS missions, the team developed a complete end-to-end workflow — from data preprocessing to deployment through an easy-to-use Streamlit app. Extensive preprocessing ensured data quality through missing-value imputation, feature selection, and encoding strategies that worked to prevent telescope-specific bias.

Light Gradient Boosted Machine (LightGBM) was identified as the optimal model following rigorous experimentation and comparison with Random Forest and CatBoost baselines. The final model achieved an overall accuracy of 83.41%, a macro-F1 score of 0.8127, and satisfactory class-wise performance in all three classes (Confirmed, Candidate, False Positive). The web interface developed is where the users can upload observational data, perform classifications in real time, and see results as numerical and graphical summaries, which makes it possible for real research scenarios.

7.2 Key Findings

The project demonstrates that one, cross-telescope model can generalize planet classification patterns despite data heterogeneity between Kepler and TESS. LightGBM's leaf-wise boosting approach delivered the best combination of accuracy, interpretability, and speed, performing at least 4% higher in overall accuracy than other models.

Feature importance analysis revealed that physically meaningful, engineered parameters — such as orbital period, transit depth, and equilibrium temperature — were most predictive, confirming that the model learned on real astrophysical relationships rather than telescope-specific trends. The introduction of a habitability scoring mechanism extended the system's scientific utility by ranking planets with greater chances of life.

7.3 Limitations

Despite the project's success, there are several limitations. The model incurred a 15–18% performance gap on cross-telescope generalization due to differences in noise properties and sampling strategies. Additionally, rare planet types (e.g., circumbinary or ultra-short-period planets) were underrepresented, and performance on those subclasses was restricted. The current model also treats each observation as static, without temporal evolution of candidate validation status across observation epochs.

7.4 Future Work

Future improvements involve reducing telescope domain shift through the utilization of domain adaptation methods such as CORAL or MMD alignment, and exploring few-shot learning approaches to enhance detection of novel planetary classes. Incorporation of temporal modeling with LSTMs or attention could also capture the path of candidate confidence across time. Incorporation of a machine learning model to classify the habitability of candidate and confirmed planets instead of sticking to astrophysical equations. Finally, incorporating an API layer and uncertainty-quantified predictions into the web system would add scientific rigor and facilitate integration with larger exoplanet discovery platforms.

In conclusion, this project demonstrates the potential for machine learning to accelerate and standardize exoplanet discovery by bridging observational gaps between missions and advancing habitability research through interpretable, automated classification.

8. References

[1] NASA Exoplanet Archive, “Kepler Objects of Interest (KOI) cumulative table,” [Online Database]. NASA Exoplanet Science Institute, California Institute of Technology, Pasadena, CA, USA. Available: https://exoplanetarchive.ipac.caltech.edu.

[2] NASA Exoplanet Archive, “TESS Project Candidates Table,” [Online Database]. NASA Exoplanet Science Institute, California Institute of Technology, Pasadena, CA, USA. Available: https://exoplanetarchive.ipac.caltech.edu.

9. Appendix

Table A-1: Parameters Used from NASA Exoplanet Archive

Column (TESS)	Alias (Kepler)	Units	Description
tfopwg_disp	koi_disposition	–	Disposition of candidate (CANDIDATE, CONFIRMED, FALSE POSITIVE)
ra	ra	degrees	Right ascension (J2000) of host star
dec	dec	degrees	Declination (J2000) of host star
pl_tranmid	koi_time0	BJD	Transit epoch (time of center of first detected transit)
pl_tranmiderr1	koi_time0_err1	BJD	$+1\sigma$ uncertainty on transit epoch
pl_tranmiderr2	koi_time0_err2	BJD	-1σ uncertainty on transit epoch
pl_orbper	koi_period	days	Orbital period of candidate
pl_orbpererr1	koi_period_err1	days	$+1\sigma$ uncertainty on orbital period
pl_orbpererr2	koi_period_err2	days	-1σ uncertainty on orbital period
pl_trandurh	koi_duration	hours	Transit duration (first to last contact)
pl_trandurherr1	koi_duration_err1	hours	$+1\sigma$ uncertainty on transit duration
pl_trandurherr2	koi_duration_err2	hours	-1σ uncertainty on transit duration
pl_trandep	koi_depth	ppm	Transit depth (flux drop during transit)
pl_trandeperr1	koi_depth_err1	ppm	$+1\sigma$ uncertainty on transit depth
pl_trandeperr2	koi_depth_err2	ppm	-1σ uncertainty on transit depth
pl_rade	koi_prad	Earth radii	Planet radius (derived from stellar radius and transit depth)
pl_radeerr1	koi_prad_err1	Earth radii	$+1\sigma$ uncertainty on planet radius

pl_radeerr2	koi_prad_err2	Earth radii	–1σ uncertainty on planet radius
pl_insol	koi_insol	Earth flux	Insolation flux relative to Earth
pl_insolerr1	koi_insol_err1	Earth flux	+1σ uncertainty on insolation
pl_insolerr2	koi_insol_err2	Earth flux	–1σ uncertainty on insolation
pl_eqt	koi_teq	K	Planet equilibrium temperature
pl_eqterr1	koi_teq_err1	K	+1σ uncertainty on equilibrium temperature
pl_eqterr2	koi_teq_err2	K	–1σ uncertainty on equilibrium temperature
st_teff	koi_steff	K	Stellar effective temperature
st_tefferr1	koi_steff_err1	K	+1σ uncertainty on stellar effective temperature
st_tefferr2	koi_steff_err2	K	–1σ uncertainty on stellar effective temperature
st_logg	koi_slogg	log10(cm/s²)	Stellar surface gravity (log g)
st_loggerr1	koi_slogg_err1	log10(cm/s²)	+1σ uncertainty on surface gravity
st_loggerr2	koi_slogg_err2	log10(cm/s²)	–1σ uncertainty on surface gravity
st_rad	koi_srad	Solar radii	Stellar radius
st_raderr1	koi_srad_err1	Solar radii	+1σ uncertainty on stellar radius